UDP Traffic and Port-Based Application Delivery

Create a UDP communication between two nodes using NS-3 and analyze the effect of link latency on end-to-end throughput.

Requirements

1. Create the topology

Create two nodes, Node1 and Node2, connected by a point-to-point link.

Node1 ——————————— Node2

P2P Link

2. Configure the first UDP application
    - Install a UdpClient on Node1.
    - Install a UdpServer on Node2.
    - Configure the client with a fixed data rate Rate1.
    - Assign an appropriate UDP port number, e.g., 5000.
3. Measure throughput

Start the UDP client and measure the end-to-end throughput while varying the point-to-point link latency.

For example:

Latency	Throughput
2 ms	_____ Mbps
5 ms	_____ Mbps
10 ms	_____ Mbps
20 ms	_____ Mbps

4. Add a second UDP application

Add another UdpClient on Node1 and another UdpServer on Node2.

Configure the second application to use a different UDP port number, for example:

Application 1 → Port 5000

Application 2 → Port 6000

Both applications use the same IP addresses, so the UDP destination port number is required to distinguish the two server applications.

5. Repeat the experiment

Repeat the throughput measurement while varying the link latency with both UDP flows active.

6. Capture and analyze packets

Enable PCAP packet tracing and examine the captured packets using Wireshark.

Verify that:

Node1 → Node2

packets with destination port 5000 are delivered to Server 1, while packets with destination port 6000 are delivered to Server 2.

7. Record observations

Prepare a table such as:

Link Latency	Flow 1 Throughput	Flow 2 Throughput
2 ms	___ Mbps	___ Mbps
5 ms	___ Mbps	___ Mbps
10 ms	___ Mbps	___ Mbps
20 ms	___ Mbps	___ Mbps